Internal Hordeolum (MONDO:0004917): A Comprehensive Disease Characterization

Summary

Internal hordeolum is a common, acute, painful, and usually self-limited suppurative infection/inflammation of a meibomian (tarsal) gland of the eyelid. It is caused by bacterial infection—classically *Staphylococcus aureus*—superimposed on obstruction and stasis of the meibomian gland orifice within the tarsal plate. It is distinguished anatomically from external hordeolum (stye), which involves the glands of Zeis or Moll associated with eyelash follicles on the anterior lamella. Internal hordeolum is an acquired, multifactorial condition of the ocular adnexa with **no known causal gene, pathogenic variant, Mendelian inheritance pattern, or dedicated genetic animal model.** Knowledge derives from aggregated disease-level clinical resources, case series, and EHR/claims-based epidemiological studies rather than from genetic patient registries.

Clinically, patients present with an acute-onset (1–3 days), typically unilateral, tender, red, swollen eyelid-margin nodule pointing toward the conjunctival surface, often accompanied by tearing and foreign-body sensation. The natural course is benign: most lesions resolve within about one week through spontaneous drainage of the abscess. However, recurrence is very common, and unresolved lesions may become chronic, evolve into a chalazion (a sterile lipogranuloma), or—rarely—spread to cause preseptal cellulitis, orbital cellulitis, or, in exceptional cases, necrotizing fasciitis. Predisposing factors center on conditions that cause meibomian gland dysfunction (MGD) and chronic lid-margin inflammation: blepharitis, ocular/acne rosacea, seborrhea, *Demodex* mite infestation, and certain systemic drugs (GLP-1/GIP receptor agonists such as tirzepatide, and CFTR modulators).

The evidence base for treatment is notably weak—repeated Cochrane systematic reviews found no reliable randomized-controlled-trial evidence for non-surgical interventions in acute internal hordeolum. In practice, management is largely conservative (warm compresses and lid hygiene), escalating to topical/systemic antibiotics for spreading infection and incision & curettage for persistent lesions. Emerging adjuncts include intense pulsed light (IPL) therapy and intralesional diluted 5-fluorouracil. Prognosis is excellent with essentially no attributable

mortality; the principal clinical burdens are recurrence, cosmetic/discomfort morbidity, and the important caveat that atypical or non-resolving lesions must be biopsied to exclude sebaceous gland carcinoma.

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1. Disease Information

Overview. Internal hordeolum is an acute, purulent inflammation of the eyelid margin caused by obstruction and bacterial infection of a meibomian (tarsal) gland. As stated in the acupuncture Cochrane review: *"When the meibomian gland in the tarsal plate is affected, internal hordeolum occurs, while when the glands of Zeis or Moll associated with eyelash follicles are affected, external hordeolum, or stye occurs"* (PMID: 28181687). It is *"a common, painful inflammation of the eyelid margin that is usually caused by a bacterial infection... The infection affects oil glands of the eyelid and can be either internal or external"* (PMID: 28068454).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0004917
MeSH	D006726 (Hordeolum)
ICD-10-CM	H00.0 (Hordeolum and other deep inflammation of eyelid); H00.03x internal hordeolum
ICD-11	9A02.1 (Hordeolum)
SNOMED CT	Hordeolum (disorder)
OMIM	Not applicable (not a Mendelian disorder)
Orphanet	Not applicable (common acquired condition, not a rare disease)

Synonyms / alternative names: internal stye, meibomian stye, acute meibomianitis / meibomian abscess, acute internal hordeolum.

Data provenance: Information derives from aggregated disease-level clinical resources (clinical reviews, Cochrane systematic reviews), case reports/series, and EHR/insurance-claims-based epidemiological studies. For example, the Japanese nationwide study used *"the National Database of Health Insurance Claims and Specific Health Checkups Open Data Japan"* (PMID: 41675090). There is no genetic patient registry for this condition.

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2. Etiology

Primary cause — infectious/mechanistic. Internal hordeolum results from bacterial infection, predominantly *Staphylococcus aureus* (MSSA, and increasingly MRSA), superimposed on obstruction of a meibomian gland orifice. In a meibomian gland dysfunction microbiome study (n=157), *"The incidence of Staphylococcus aureus was higher than anticipated across all severity groups"* (PMID: 27473509). Importantly, the underlying chronic gland disease is often not primarily infectious: *"Primary meibomitis appears not to be a primarily infectious entity but to represent a facet of generalized sebaceous gland dysfunction and to be found in association with seborrheic dermatitis or acne rosacea"* (PMID: 6218459).

Genetic risk factors. None established. There is no causal gene, susceptibility locus, chromosomal abnormality, or Mendelian inheritance pattern described for internal hordeolum. It is a multifactorial, acquired condition.

Environmental / acquired risk factors. - **Blepharitis and meibomian gland dysfunction (MGD):** Hordeolum and chalazion *"can occur in association with underlying diseases causing Meibomian gland dysfunction and/or chronic blepharitis"* (PMID: 34379160). - **Ocular/acne rosacea and seborrhea:** generalized sebaceous gland dysfunction (PMID: 6218459); ocular rosacea can present with *"multiple recurrent chalazion on both eyelids"* with immunohistochemistry *"positive for LL-37"* (PMID: 38898849). Blepharitis dominates pediatric ocular rosacea: *"Blepharitis was the most common eyelid manifestation (98.4%)"* (PMID: 42757848). - ***Demodex* infestation** (see Section 5). - **Systemic drugs:** GLP-1/GIP receptor agonists (tirzepatide) and CFTR modulators (elexacaftor-tezacaftor-ivacaftor) — an emerging iatrogenic trigger via meibomian gland obstruction (PMID: 42756500; PMID: 41967150). - **Age/sex:** bimodal age peaks; slight female predominance in youth (see Section 9).

Protective factors. No genetic protective variants are described. Environmental/behavioral protective measures are essentially preventive hygiene: lid hygiene, warm compresses, and treatment of underlying MGD/rosacea reduce recurrence frequency; IPL reduces hordeolum frequency (PMID: 34629862).

Gene–environment interactions. Not applicable in a Mendelian sense. The relevant "host–environment" interaction is the interplay between host sebaceous-gland biology (rosacea/seborrhea/innate immune dysregulation via cathelicidin LL-37), commensal/pathogenic bacteria (*S. aureus*), ectoparasites (*Demodex*), and gland-altering systemic drugs.

3. Phenotypes

Core clinical phenotype: An acute-onset, usually unilateral, painful, tender, red, swollen localized eyelid-margin nodule (abscess) pointing toward the conjunctival (inner) surface. Patients present *"with symptoms such as redness, swelling, tearing, itchiness, or a foreign body sensation"* (PMID: 32622681), and the lesion is fundamentally *"a common, painful inflammation of the eyelid margin"* (PMID: 28068454).

Phenotype	Type	HPO suggestion	Characteristics
Painful eyelid nodule/swelling	Symptom + physical sign	HP:0000498 (Abnormality of the eyelid)	Acute onset (1–3 d); mild–moderate; localized
Eyelid erythema	Clinical sign	HP (eyelid erythema)	Acute; resolves ~1 wk
Eyelid edema/swelling	Clinical sign	HP:0100539 (Periorbital edema)-adjacent	Localized, unilateral
Epiphora (tearing)	Symptom	HP:0009926 (Epiphora)	Mild
Foreign-body sensation / ocular discomfort	Symptom	HP (ocular discomfort)	Mild–moderate
Eyelid pain/tenderness	Symptom	HP:0200026 (Ocular pain)-adjacent	Moderate; localized

Onset: any age, acute (see Section 8). **Severity:** typically mild–moderate; occasionally severe with spread. **Progression:** episodic/self-limited (~1 week), frequently recurrent. **Frequency among affected:** pain and eyelid-margin swelling are essentially universal defining features; tearing, itch, and foreign-body sensation are common.

Quality of life impact: Generally modest and transient; discomfort, cosmetic concern, and interference with vision if large. Chalazion/hordeolum-spectrum lesions can induce corneal astigmatism and higher-order aberrations when sizeable, affecting vision (PMID: 41884092). IPL studies used the Ocular Surface Disease Index (OSDI) as a QoL/symptom instrument (PMID: 41731784).

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4. Genetic/Molecular Information

Not applicable. Internal hordeolum is an acquired infectious/inflammatory condition with **no causal genes, no pathogenic variants, no modifier genes, no established disease-specific epigenetic signature, and no chromosomal abnormalities.** There is no OMIM entry. There are no ClinVar/HGMD variants, no gnomAD allele frequencies, and no somatic/germline mutation profile associated with the disease itself.

The only molecularly-characterized host contributor identified is the innate-immune peptide **cathelicidin LL-37** in the setting of rosacea-associated recurrent eyelid lesions: immunohistochemistry was *"positive for LL-37"* (PMID: 38898849). This is a downstream inflammatory mediator of the predisposing rosacea diathesis, not a causal genetic variant of hordeolum.

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5. Environmental Information

Environmental factors. No toxins, radiation, or occupational exposures are established as direct causes. The dominant modifiable exposures are pharmacological (see below) and microbial.

Lifestyle factors. Lid hygiene practices modulate risk/recurrence. Poor lid hygiene and chronic lid-margin inflammation predispose; conversely, warm compresses and lid hygiene are preventive.

Infectious agents. - ***Staphylococcus aureus*** — the principal bacterial pathogen (NCBI Taxonomy: *Staphylococcus aureus*, txid1280); MSSA and MRSA. Elevated on lid margins in MGD (PMID: 27473509). Bacterial lipases contribute to inflammation: *"the production of bacterial lypolytic exoenzymes that split neutral lipids, resulting in an increased level of free fatty acids in the tears"* (PMID: 6218459). - ***Demodex folliculorum*** **and *Demodex brevis** *(ectoparasitic*

mites) — commonly infest eyelids and drive blepharitis/MGD that predisposes to hordeolum. In squamous blepharitis, "The overall Demodex detection rate was 92% (92/100) in this study" *(PMID: 42426675), and* "In Demodex blepharitis, abnormal or reduced lipid production from the Zeis and/or meibomian glands may contribute to DED"* (PMID: 42387805).

Pharmacological environmental triggers (iatrogenic). Tirzepatide (dual GLP-1/GIP receptor agonist) caused simultaneous/recurrent internal hordeola with a positive dechallenge/rechallenge: *"lesions lessened after tirzepatide discontinuation and worsened upon rechallenge, a positive rechallenge providing strong evidence of causality"* and *"Both patients lacked classic infectious blepharitis signs despite active lesions, supporting a sterile, obstructive meibomian gland process rather than infection"* (PMID: 42756500). CFTR modulators are similarly implicated in children with cystic fibrosis (PMID: 41967150).

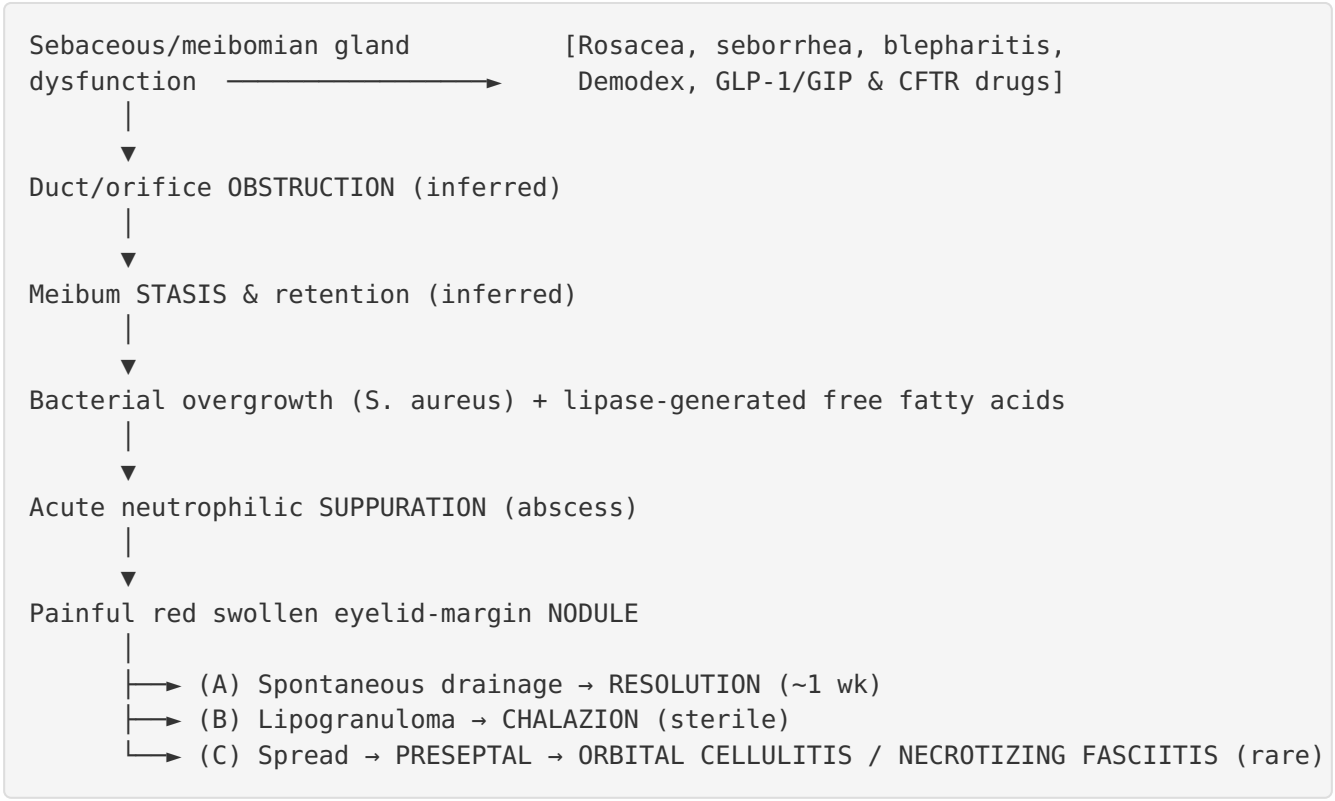
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6. Mechanism / Pathophysiology

Ordered causal chain

1. **Predisposing sebaceous/meibomian gland dysfunction** (blepharitis, ocular/acne rosacea, seborrhea, *Demodex*, or altered meibum from systemic drugs) **leads to →**
2. **Obstruction of the meibomian gland duct/orifice** within the tarsal plate *(inferred from meibomian gland biology)* **which results in →**
3. **Stasis and retention of sebaceous secretions (meibum)** *(inferred)* **which promotes →**
4. **Bacterial overgrowth/infection**, predominantly *Staphylococcus aureus*, whose lipases split neutral lipids into pro-inflammatory free fatty acids *(demonstrated by microbiology/biochemistry)* **which triggers →**
5. **Acute neutrophilic suppurative inflammation** (abscess/pus formation) within the gland *(demonstrated histopathologically)* **which produces →**
6. **The clinical painful, red, swollen eyelid-margin nodule.**

Branches from step 5/6: - **Branch A (resolution):** spontaneous drainage → resolution in ~1 week. - **Branch B (chronic/sterile):** unresolved lipogranulomatous organization → chalazion (sterile chronic granuloma). - **Branch C (spread):** extension → preseptal cellulitis → (rarely) orbital cellulitis / necrotizing fasciitis.



Supporting detail

- **Initiating lesion:** *"Hordeolum is an acute, purulent inflammation of the eyelid margin usually caused by obstructed orifices of the sebaceous glands of the eyelid"* (PMID: 28181687).
- **Biochemical inflammatory step:** bacterial lipolytic exoenzymes generate irritant free fatty acids (PMID: 6218459).
- **Sterile-obstructive upstream branch:** distinct non-infectious mechanism supported by drug-induced cases *"supporting a sterile, obstructive meibomian gland process rather than infection"* (PMID: 42756500).
- **Innate-immune amplification (rosacea):** cathelicidin **LL-37** (PMID: 38898849).

Steps 2–3 are mechanistically inferred from meibomian gland biology; **steps 4–6** are demonstrated by microbiology and histopathology.

Molecular pathways / cellular processes: acute inflammation, neutrophil recruitment and degranulation, innate immune antimicrobial peptide (cathelicidin) activation, lipid metabolism disturbance (holocrine sebaceous secretion). Granulomatous organization occurs if the lesion becomes chronic (chalazion).

Suggested ontology terms. - GO biological processes: GO:0006954 (inflammatory response); GO:0006909 (phagocytosis); GO:0042742 (defense response to bacterium); GO:0006629 (lipid metabolic process). - CL cell types: CL:0000775 (neutrophil); CL:0000235 (macrophage); sebaceous gland cell / meibocyte; CL:0000738 (leukocyte). - CHEBI: CHEBI:35366 (fatty acid) / free fatty acids; neutral lipids/triglycerides.

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7. Anatomical Structures Affected

Organ/tissue level. The primary structure is the **meibomian (tarsal) gland**, a holocrine sebaceous gland embedded in the **tarsal plate** of the eyelid, on the **posterior lamella**, with the lesion pointing toward the **conjunctival (inner) surface**: *"When the meibomian gland in the tarsal plate is affected, internal hordeolum occurs"* (PMID: 28181687). This contrasts with external hordeolum, which affects glands of Zeis/Moll on the anterior lamella at the lash line.

Body system: visual/ocular adnexal system (eyelid). **Laterality:** usually **unilateral** and localized. **Secondary involvement:** periorbital soft tissue (preseptal space) and, rarely, the orbit; hordeolum is a leading predisposing factor for preseptal cellulitis (15.5–24.8% of cases) (PMID: 34607790).

Tissue and cell level: glandular (sebaceous) epithelium of the meibomian gland (meibocytes); infiltrating neutrophils and, in chronic phase, macrophages/lipogranuloma.

Subcellular level: holocrine secretion involves whole-cell disintegration; relevant GO cellular components include lipid droplet (GO:0005811) and the secretory apparatus.

Suggested ontology terms. UBERON: UBERON:0001711 (eyelid); UBERON:0004841 (tarsal gland / meibomian gland); UBERON:0001801 (palpebral conjunctiva). CL: meibocyte/sebaceous gland cell.

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8. Temporal Development

Onset. Acute, developing over ~1–3 days, at **any age**, with bimodal peaks in childhood/adolescence (10–14 y) and mid-adulthood (30–39 y): *"Invasive treatments for hordeolum showed bimodal peaks in males (age 10-14 years: 82.9; age 35-39 years: 72.1) and females (age 10-14 years: 102.4; age 30-34 years: 78.3)"* (PMID: 41675090). There is no congenital form.

Progression / course. Short and self-limited: *"The onset of hordeolum is usually self limited, and may resolve in about a week with spontaneous drainage of the abscess. When the condition is severe, it can spread to adjacent glands and tissues. Recurrences are very common"* (PMID: 28181687). The course is **episodic/recurrent** rather than steadily progressive. There is no staged progression system. Unresolved lesions transition to chronic chalazion.

Patterns. Remission is typically **spontaneous** (drainage) or treatment-facilitated. The main "critical period" for intervention is early warm-compress/lid-hygiene use and, for spreading disease, prompt antibiotics to prevent cellulitis.

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9. Inheritance and Population

Epidemiology. Internal hordeolum is common. Japanese nationwide claims data (2014–2022, 638,699 invasive hordeolum procedures) reported an average invasive treatment rate of **56.2 per 100,000 person-years**: *"The annual average rates were 40.9 for ITC and 56.2 for ITH per 100 000 person-years. Sex differences were prominent in younger groups: the overall female-to-male ratio was 1.1:1 for ITC and 1.0:1 for ITH, with a greater female predominance in youth"* (PMID: 41675090). Note that invasive-treatment rates capture only the subset requiring procedures; true incidence (including conservatively-managed cases) is higher. In a Korean military teleophthalmology cohort, hordeolum/chalazion together were the single most common diagnosis: *"The most common diagnoses were hordeolum/chalazion (39.3%) and viral conjunctivitis (20.3%)"* (PMID: 42299138).

Genetic etiology. Not applicable — no inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity role, or carrier frequency, because there is no causal gene.

Population demographics. - **Sex ratio:** approximately 1.0:1 overall for hordeolum, with a greater female predominance in youth (PMID: 41675090). - **Age distribution:** bimodal (10–14 y and 30–39 y). - **Geographic distribution:** cosmopolitan; no endemic pattern. No population-specific variants (none exist).

Metric	Value	Source
Invasive-treatment rate (hordeolum)	56.2 / 100,000 person-years	PMID: 41675090
Female:male (overall)	~1.0:1	PMID: 41675090
Male age peaks	10–14 y (82.9); 35–39 y (72.1)	PMID: 41675090
Female age peaks	10–14 y (102.4); 30–34 y (78.3)	PMID: 41675090
Share of ophthalmic consults (hordeolum/chalazion)	39.3%	PMID: 42299138

10. Diagnostics

Clinical diagnosis is standard. *"Due to the typical morphological features and clinical course, the diagnosis can mostly be clinically established"* (PMID: 34379160). Diagnosis rests on inspection of the eyelid margin revealing a tender, localized, erythematous nodule.

Laboratory tests / microbiology: not routinely required; culture (e.g., for *S. aureus*/MRSA) is reserved for severe, atypical, or recurrent cases or when complications supervene.

Biopsy / histopathology — critical safety rule: Persistent, recurrent, or clinically atypical lesions require excisional biopsy to exclude malignancy, especially **sebaceous gland carcinoma**: *"In persistent, recurrent or clinically atypical cases malignant tumors must be excluded as a differential diagnosis by excisional biopsy and histopathological assessment"* (PMID: 34379160). Real cases of malignancy masquerading as hordeolum include metastatic invasive lobular breast carcinoma—*"She was initially managed for presumed hordeolum and preseptal cellulitis without response over 4 months"* (PMID: 42552075)—and apocrine/sebaceous carcinomas (PMID: 40531581).

Genetic / omics diagnostics: Not applicable — no genetic testing, gene panels, karyotyping, CMA, or disease-specific transcriptomic/proteomic/metabolomic signature.

Differential diagnosis: chalazion (chronic, non-tender granuloma), external hordeolum (stye at lash line), preseptal/orbital cellulitis, sebaceous gland carcinoma, apocrine carcinoma of the glands of Moll, molluscum, and other eyelid tumors. In children, eyelid cutaneous horns are frequently *"preceded by a hordeolum or chalazion"* and follow a benign course (PMID: 39393420).

Screening: No population screening applies (not a genetic or preventable-by-screening disease).

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11. Outcome/Prognosis

Excellent prognosis. Internal hordeolum is benign and self-limited, usually resolving in ~1 week with spontaneous drainage (PMID: 28181687). **Mortality is essentially nil**, occurring only via exceedingly rare complications such as necrotizing fasciitis: *"We report an unusual case of periorbital NF caused by methicillin-sensitive Staphylococcus aureus (MSSA) in a previously healthy 48-year-old woman, originating from a neglected, recurrent hordeolum-an otherwise benign eyelid condition"* (PMID: 41787906).

Morbidity. The main burdens are **recurrence** (very common), transient discomfort/cosmetic effect, and chronic sequelae. Unresolved lesions evolve into chalazion, which can cause **meibomian gland loss/dropout**: a pediatric chalazion cohort demonstrated *"Chalazion can lead to meibomian gland loss"* with significant gland dropout versus controls (meiboscores 0.78–0.89 vs 0.40, P<.05) (PMID: 42116414). Larger lesions can transiently induce corneal astigmatism/higher-order aberrations (PMID: 41884092).

Complications: chalazion; preseptal cellulitis (hordeolum accounts for 15.5–24.8% of adult preseptal cellulitis predisposing factors, PMID: 34607790); rarely orbital cellulitis and necrotizing fasciitis.

Prognostic factors: severity, promptness of drainage/treatment, and control of underlying MGD/rosacea/Demodex determine recurrence risk. No molecular prognostic biomarkers exist.

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12. Treatment

Evidence caveat. The treatment evidence base is weak. The Cochrane review's stated objective was *"to investigate the effectiveness, and when possible, the safety, of non-surgical treatments for acute internal hordeola compared with observation or placebo"* (PMID: 28068454) — and no eligible randomized trials were found. Acupuncture is likewise unproven: *"it is unclear if acupuncture is an effective and safe treatment for acute hordeolum"* (PMID: 28181687).

Treatment ladder.

Tier	Intervention	Evidence / notes	NCIT suggestion
First-line	Warm compresses; lid hygiene/scrubs	Standard conservative care; most resolve spontaneously. Warming masks raise eyelid temperature to therapeutic range (PMID: 42361717)	NCIT:C15277 (Supportive Care)
Second-line	Topical/systemic antibiotics	For spreading infection (anti-staphylococcal)	NCIT:C258 (Antibiotic)
Procedural	Incision & curettage (I&C)	For persistent lesions	NCIT (surgery/ incision-drainage)
Emerging	Intense pulsed light (IPL)	Prospective cohort: *"the IPL group exhibited a statistically significant reduction in OSDI scores following treatment"* (PMID: 41731784); BBL-IPL *"has shown to reduce hordeolum and blepharitis frequency"* (PMID: 34629862)	NCIT (phototherapy)
Emerging	Intralesional diluted 5-fluorouracil	*"Local injection of 6-fold diluted 5-fluorouracil may represent a potential minimally invasive option for selected patients with persistent eyelid inflammatory nodules"* (PMID: 42367702)	NCIT:C505 (Fluorouracil)
Underlying disease	Oral doxycycline/ azithromycin (rosacea); lotilaner (Demodex)	Rosacea chalazia resolved with *"oral administration of doxycycline and azithromycin"* (PMID: 38898849); lotilaner 0.25% for Demodex blepharitis (PMID: 42371506)	NCIT:C303 (Doxycycline)
Iatrogenic	Discontinue/adjust offending drug	Positive dechallenge for tirzepatide (PMID: 42756500)	—

Pediatric procedural management (chalazion I&C) under general anesthesia is common and safe (n=4,616 series; chalazion I&C = 62.89% of pediatric ophthalmic day surgery, PMID: 39115475).

Pharmacogenomics / advanced (gene/cell/RNA) therapies: Not applicable.

13. Prevention

- **Primary prevention:** lid hygiene, warm compresses, control of blepharitis/MGD; avoiding/managing gland-obstructing systemic drugs where feasible.
- **Secondary prevention:** early warm compress/lid hygiene at symptom onset; prompt antibiotics if spreading to avert cellulitis.
- **Tertiary prevention:** treat underlying rosacea (doxycycline/azithromycin, PMID: 38898849) and *Demodex* (lotilaner, PMID: 42371506); IPL reduces recurrence frequency (PMID: 34629862).
- **Immunization / genetic screening / counseling:** Not applicable (no vaccine, no genetic basis).
- **Public health:** general eye-hygiene education.

14. Other Species / Natural Disease

- **Taxonomy of affected host:** *Homo sapiens* (NCBI:txid9606). The causative pathogen is *Staphylococcus aureus* (NCBI:txid1280); ectoparasites *Demodex folliculorum*/*D. brevis*.
- **Comparative biology:** Meibomian (tarsal) glands are **holocrine sebaceous glands conserved across mammals**, so an anatomical substrate exists in animals. However, **no established rodent/zebrafish/Drosophila genetic model of "internal hordeolum" exists**, and formal veterinary "hordeolum" literature is minimal. Analogous eyelid gland conditions in animals are described as chalazia/meibomian adenitis rather than "internal hordeolum."
- **Zoonosis:** Not applicable — hordeolum itself is not transmissible; *S. aureus* can colonize multiple hosts but the disease is not classically zoonotic.

15. Model Organisms

Not applicable / none established. There is no dedicated genetic or induced animal model of internal hordeolum (no knockout, knock-in, transgenic, conditional, or humanized model). Related meibomian gland dysfunction can be studied in mammalian models of MGD/dry eye, but these do not specifically recapitulate the acute suppurative internal hordeolum phenotype. Consequently, phenotype recapitulation, model limitations, and model databases (MGI, RGD, ZFIN, etc.) are **not applicable** for this specific disease entity.

Mechanistic Model / Interpretation

Internal hordeolum is best understood as the **acute suppurative endpoint of a chronically primed eyelid oil-gland system**. The upstream soil is meibomian gland dysfunction—driven by rosacea/seborrhea (with innate-immune LL-37 involvement), *Demodex* infestation, or drug-altered meibum. This soil predisposes to duct obstruction and meibum stasis (inferred steps), which provides a lipid-rich niche for *S. aureus*, whose lipases liberate pro-inflammatory free fatty acids. The resulting neutrophilic abscess is the clinically visible lesion. The system has three exits: spontaneous drainage (the common, benign outcome), sterile granulomatous organization into a chalazion, or bacterial spread into the preseptal/orbital compartments. This model explains the disease's key clinical features: its acuity (infection), its self-limitation (drainage), its recurrence (persistent underlying MGD soil), and its therapeutic logic (treat the abscess acutely, but prevent recurrence by remediating the gland dysfunction).

The **"two-hit" framing**—chronic obstructive/sterile predisposition plus acute bacterial infection—reconciles two literatures that might otherwise seem contradictory: the observation that primary meibomitis is "not a primarily infectious entity" (PMID: 6218459) and drug-induced "sterile obstructive" hordeola (PMID: 42756500) on one hand, versus the demonstrable role of *S. aureus* on the other.

Evidence Base

PMID	Role in this report	Type
28181687	Core definition, anatomy, self-limited course, obstruction mechanism	Cochrane review
28068454 / 20824865	Definition; treatment evidence gap	Cochrane review
6218459	Sebaceous dysfunction etiology; bacterial lipase → free fatty acids	Clinical/mechanistic
27473509	Elevated *S. aureus* in MGD	Microbiome study
41675090	Epidemiology, incidence, bimodal age, sex ratio	Nationwide claims
42299138	High prevalence in real-world cohort	Teleophthalmology review
34379160	Clinical diagnosis; biopsy for atypical/recurrent; MGD/blepharitis risk	Clinical review
42756500	Tirzepatide-induced sterile obstructive hordeola; rechallenge causality	Case series
41967150	CFTR modulator association	Case report
38898849	Rosacea, LL-37; doxycycline/azithromycin treatment	Case report
42757848	Pediatric ocular rosacea, blepharitis 98.4%	Case series
42426675	*Demodex* 92% in blepharitis	Clinical study
42387805	*Demodex* → Zeis/meibomian lipid dysfunction	Clinical study
41787906	Necrotizing fasciitis complication	Case report
34607790	Hordeolum as preseptal cellulitis predisposing factor	Retrospective
42116414	Chalazion → meibomian gland loss	Observational
42552075	Malignancy masquerading as hordeolum	Case report
41731784	IPL improves OSDI	Prospective cohort
34629862	BBL-IPL reduces hordeolum frequency	Clinical study

PMID	Role in this report	Type
42367702	Intralesional 5-FU	Case series
42371506	Lotilaner for Demodex blepharitis	Case report
32622681	Presenting symptoms of eyelid infection	Clinical review
42361717	Warming mask eyelid temperature	Experimental
39115475	Pediatric I&C safety	Retrospective
39393420	Cutaneous horn preceded by hordeolum/chalazion	Case series
41884092	Lesion-induced astigmatism/HOAs	Review

Limitations and Knowledge Gaps

1. **No randomized-controlled-trial evidence** exists for non-surgical treatment of acute internal hordeolum; practice is guided by expert opinion and observational data (PMID: 28068454).
2. **Incidence figures are procedure-based** (invasive treatments), so the 56.2/100,000 rate underestimates true incidence, which includes conservatively-managed and self-resolving cases.
3. **Mechanistic steps 2–3 (obstruction, stasis) are inferred** from meibomian gland biology rather than directly demonstrated in hordeolum tissue.
4. **No molecular/omics characterization** specific to internal hordeolum (no transcriptomic, proteomic, metabolomic, or single-cell studies of the lesion itself).
5. **No animal model**, limiting mechanistic dissection.
6. **Emerging drug associations** (GLP-1/GIP agonists, CFTR modulators) rest on small case series; magnitude of risk is unquantified.
7. **IPL and intralesional 5-FU evidence** comes from small, mostly uncontrolled or single-center studies.

Proposed Follow-up Experiments / Actions

1. **Conduct a pragmatic RCT** of warm compress ± topical antibiotic vs. observation for acute internal hordeolum, using standardized outcomes (time-to-resolution, recurrence, OSDI) to close the Cochrane-identified evidence gap.
2. **Pharmacovigilance analysis** (FAERS/pharmacoepidemiology) to quantify hordeolum/chalazion risk with GLP-1/GIP agonists and CFTR modulators, establishing incidence and dose relationship.
3. **Microbiome + host-response profiling** of drained hordeolum contents (16S/metagenomics + neutrophil/LL-37 markers) to confirm the two-hit infectious/sterile model and stratify infectious vs. sterile lesions.
4. **Prospective imaging cohort** (meibography) to test whether recurrent hordeola predict long-term meibomian gland dropout, extending the pediatric chalazion findings (PMID: 42116414).
5. **Controlled trial of IPL** for recurrence prevention in high-risk (rosacea/MGD/Demodex) patients, with meibomian gland function endpoints.
6. **Prospective study of biopsy yield** in non-resolving eyelid nodules to refine the threshold/timing for excluding sebaceous carcinoma.

Report compiled from 16 confirmed findings across 5 investigation iterations and 34 reviewed papers. Evidence types span human clinical reviews, Cochrane systematic reviews, nationwide claims/EHR studies, case series, and case reports. No model-organism, in vitro, or computational disease-specific data were available for this entity.
